



Station-keeping performance analysis for high altitude balloon with altitude control system

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ARTICLE INFO

Article history:

Received 7 May 2019

Received in revised form 21 June 2019

Accepted 21 June 2019

Available online 26 June 2019

Keywords:

Altitude control system

Flight trajectory

High altitude balloon

Station-keeping

Wind field

ABSTRACT

Station-keeping endurance of high altitude balloon is the foundation of the assignments of environmental monitoring and communicational relaying. Exploiting the natural wind-field that varies with altitude by the altitude control system to extend the station-keeping endurance is proposed in this paper. A Matlab program is developed based on the theoretical model to simulate the station-keeping performance of the balloon in the real wind field. The trajectories of the balloon in different wind fields and the states of the balloon caused by the venting and pumping processes are discussed in detail. The results show that with the altitude control system it is possible to retain the balloon within the designated district for few days to a week. This can serve as a guideline for the design and initial flight tests of the serviceable high altitude balloon.

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1. Introduction

The stratosphere, with the characteristic of stable atmosphere and weakly vertical convection, is the ideal location to deploy aerial platforms for the purpose of surveillance, early alarming, communications relay, navigation and environmental monitoring [1]. Many countries such as America, Japan, and European countries, have developed many investigations and experiments of high altitude platforms.

With the breakthrough of film materials and optimum design technologies [2–4], the high altitude balloon has become one of the most important platforms for the research and development of stratosphere. However, different with the satellite that can be controlled, the balloon will drift freely with the prevailing winds [5–7]. As a result, the trajectory tracking and control of the balloon are very difficult, which limits the further application. In previous research [8], we investigated the trajectory of the free balloon in the real wind. The result shows that the traveled horizontal distance between the balloon and the launch site within 24 hours varies from dozens to hundreds of kilometers. The experiments developed by NASA [9,10] also proved that the balloon cannot keep the station to provide persistent surveillance without control.

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Station-keeping endurance is indispensable for the balloon to accomplish the observation and telecommunication mission [11,12]. The propulsion system is employed for the airship to overcome the wind speed and maintain the location [13,14]. However, it is not suitable for the balloon. Considering the phenomenon that wind directions vary with altitude [15], some brilliant methods that exploits the natural wind field are proposed to control the station of the balloon.

Nock [16] developed a winged balloon guidance system (BGS) that exploits the natural wind field that varies with the altitude to generate passive lateral control forces on a balloon using a tether-deployed aerodynamic surface several kilometers below the balloon. The capability of the BGS in control implementation was demonstrated. The research confirmed that, combined with available differential winds, the BGS can provide a very significant balloon guidance capability for long endurance high altitude balloon. Yoder [17] analyzed the performance of a tether and sail-based trajectory control system for extra-terrestrial scientific balloon missions. A proportional integral (PI) controller was designed to demonstrate rudimentary control of the balloon system for maintaining and changing latitudes in the presence of favorable and adverse winds. Ramesh [18] presented the dual-balloon and balloon-stratosail systems that have a control device tethered to the main balloon to retain the balloon system within a specific region. The simulation results demonstrate that both the systems can retain the trajectories within the district with radius of few hundred kilometers.

Adjusting the float altitude of the balloon to utilize the winds with different directions is also an effective approach to keep station. There are two methods to adjust the float altitude, changing the buoyancy and changing the mass. The buoyancy of the balloon can be changed by varying the volume. Voss [19] developed the volume control systems called mechanical compression (MC) and differential expansion (DE). By drawing the base and apex fittings together, the volume can be reduced by as much as 50%. Athar [20] installed the heating mechanism at the top of the balloon to heat the lift gas inner balloon and then change the volume. The most successful and widely known system to change the mass of the balloon is the air ballast. The balloon with air ballast evolved rapidly since the early 1990s and have been successfully deployed on a number of major studies. Smith [21] gave two air ballast methods known as the anchor balloon method and the ballonet method to adjust the altitude. Air is released from the constant volume balloon for ascent while air is pumped into to add mass to the system for descent. Hall [22] discussed the underlying theory governing the design of the balloon with air ballast. The mass and the energy consumption to achieve the given altitude change were calculated. In addition, Google's Project Loon also employed the air ballast methods to achieve the altitude control over small ranges [23,24]. However, very few investigations have been conducted on the detailed station-keeping strategy and the station-keeping performance up to now.

In this paper, in order to give a detailed analysis on the station-keeping performance of the high altitude balloon with the altitude control system (ACS), a Matlab program is developed based on the models of real wind field model, thermal model, dynamic model and the station-keeping model. The trajectories of the balloon with the ACS in different wind fields are simulated. The result shows that the balloon can be retained within the desired district, which provides strong foundation for the design and flight experiment of the serviceable high altitude balloon.

2. Altitude control system

Fig. 1 shows the schematic diagram of the station-keeping method for the balloon. To fix the balloon within the designated district and extend the station-keeping endurance, we can reverse the cruise direction before the balloon racks out of the district. Considering that the real wind direction varies with altitude, it is feasible to change the direction by altering the float altitude after identifying the wind layer with the desired direction. This can be achieved by employing the ACS.

Fig. 2 shows the principle diagram of the ACS. The balloon is composed of main Helium bag and air ballonet that are separated by a membrane. The main Helium bag filled with Helium can provide buoyancy to balance the weight of the balloon while the air ballonet filled with air can adjust the weight by opening the exhaust valve or aeration valve to vent or pump the air. Normally, the Helium bag has no mass transfer with the external atmosphere. To descend the balloon, the air in the atmosphere will be compressed into the air ballonet by a compressor to increase the weight. The minimum altitude ($h_{\min}$) can be achieved when

$$P_{in}(h) - P_{atm}(h) = \Delta P_{\max} \quad (1)$$

where $P_{in}(h)$ and $P_{atm}(h)$ are the pressure of inner gas and atmosphere at the altitude of h . $\Delta P_{\max}$ is the maximum differential pressure that the film can withstand. By contrast, the air will be exhausted to reduce the weight and then ascend the balloon. The maximum altitude ($h_{\max}$) can be reached when

$$P_{in}(h) = P_{atm}(h) \quad (2)$$

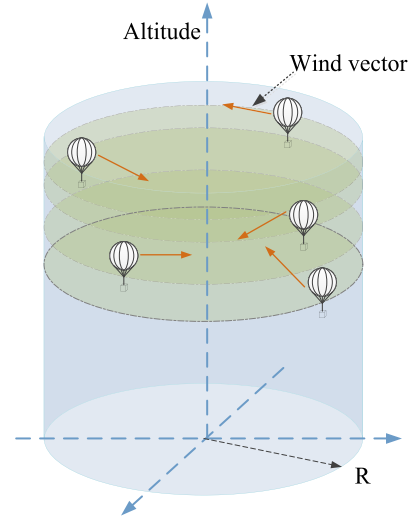


Fig. 1. Schematic of the station-keeping mechanism of the balloon.

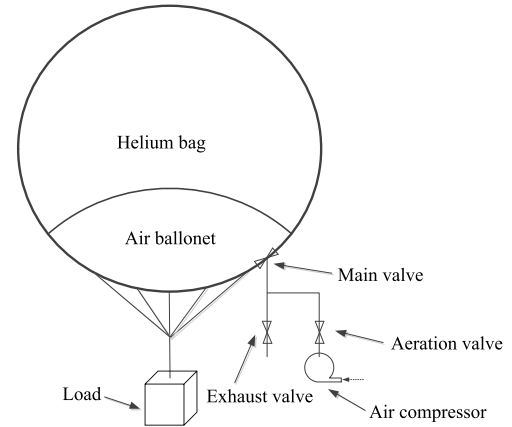


Fig. 2. Sketch of the altitude adjustment mechanism of the balloon.

As a result, the altitude can be changed in the range from $h_{\min}$ to $h_{\max}$. Generally, the altitude difference between $h_{\min}$ and $h_{\max}$ is about one or several kilometers. Combined with different wind directions at different altitudes and effective control strategies, station-keeping can be attained with the ACS.

3. Theoretical model

To simulate the station-keeping performance of the balloon, the fundamental assumptions made in this paper are as follows.

- (1) The variations of the atmosphere temperature, pressure and density with altitude follow the US Standard Atmosphere Model (U.S. Standard Atmosphere, 1976).
- (2) The Helium and the air in the balloon are assumed to follow the idea gas law.
- (3) The density, temperature and pressure of the Helium and the air are considered uniform inside the balloon.
- (4) The internal Helium pressure is presumed to be the same as the internal air pressure.
- (5) The turbulent wind in stratosphere is ignored. The wind fields are steady within several days [25–27].

3.1. Wind field model

The wind fields of Changsha (28°N, 112°E) on May 15, June 15 and July 15 in 2018 are employed in this paper. The wind profiles ranging from 18 km to 24 km are measured by radiosonde balloons and presented in Fig. 3. Generally, the wind conditions in stratosphere fluctuate slightly compared to that in troposphere. Therefore, we assume that the wind profiles are steady within several days for simplicity. With fifth-term Fourier series, the east and north components of the wind fields are fitted to the measured data.

$$v_{E/N} = a_{0,E/N} + A_{E/N} \begin{bmatrix} \cos(\zeta_{E/N} \cdot h/h_c) \\ \cos(2\zeta_{E/N} \cdot h/h_c) \\ \cos(3\zeta_{E/N} \cdot h/h_c) \\ \cos(4\zeta_{E/N} \cdot h/h_c) \\ \cos(5\zeta_{E/N} \cdot h/h_c) \end{bmatrix} + B_{E/N} \begin{bmatrix} \sin(\zeta_{E/N} \cdot h/h_c) \\ \sin(2\zeta_{E/N} \cdot h/h_c) \\ \sin(3\zeta_{E/N} \cdot h/h_c) \\ \sin(4\zeta_{E/N} \cdot h/h_c) \\ \sin(5\zeta_{E/N} \cdot h/h_c) \end{bmatrix} \quad (3)$$

where h_c is a dimensional scaling parameter, $h_c = 20000$. E and N are the easterly wind and northerly wind respectively.

The coefficients are given as follows:

May 15, 2018:

$$\begin{cases} a_{0,E,May} = 2.285 \\ A_{E,May} = [1.248 \quad 1.418 \quad -0.3716 \quad -1.124 \quad 0.97] \\ B_{E,May} = [-2.805 \quad 0.1873 \quad 1.578 \quad -0.1496 \quad 0.5454] \\ \zeta_{E,May} = 15.71 \end{cases} \quad (4)$$

$$\begin{cases} a_{0,N,May} = 0.7142 \\ A_{N,May} = [-0.2775 \quad 0.3651 \quad 0.9881 \quad -1.239 \quad 0.7975] \\ B_{N,May} = [0.1488 \quad -1.161 \quad -0.921 \quad 1.538 \quad 0.6094] \\ \zeta_{N,May} = 17.24 \end{cases} \quad (5)$$

June 15, 2018:

$$\begin{cases} a_{0,E,Jun} = -1.547 \\ A_{E,Jun} = [6.938 \quad 6.641 \quad -2.954 \quad -0.3597 \quad -0.3973] \\ B_{E,Jun} = [-4.467 \quad 2.42 \quad 1.254 \quad -1.226 \quad -1.436] \\ \zeta_{E,Jun} = 15.91 \end{cases} \quad (6)$$

$$\begin{cases} a_{0,N,Jun} = 1.156 \\ A_{N,Jun} = [1.056 \quad 2.673 \quad 0.3392 \quad -2.429 \quad -2.564] \\ B_{N,Jun} = [2.745 \quad 2.619 \quad 0.5838 \quad -0.7461 \quad -0.6763] \\ \zeta_{N,Jun} = 13.9 \end{cases} \quad (7)$$

July 15, 2018:

$$\begin{cases} a_{0,E,Jul} = 12.52 \\ A_{E,Jul} = [2.101 \quad 0.2114 \quad 1.836 \quad 0.3126 \quad -0.4203] \\ B_{E,Jul} = [-0.4693 \quad -1.851 \quad 2.057 \quad 0.1317 \quad 0.7141] \\ \zeta_{E,Jul} = 15.53 \end{cases} \quad (8)$$

$$\begin{cases} a_{0,N,Jul} = 1.286 \\ A_{N,Jul} = [-1.263 \quad -0.8208 \quad -1.236 \quad 0.04674 \quad 0.6516] \\ B_{N,Jul} = [1.808 \quad -0.7048 \quad 0.5374 \quad 0.05691 \quad -0.6128] \\ \zeta_{N,Jul} = 16.58 \end{cases} \quad (9)$$

3.2. Thermal model

The thermal environment of high altitude balloon can be divided into external and internal thermal environment. The external environment mainly comprises direct solar radiation, atmosphere scattered radiation, earth reflected radiation, atmosphere infrared radiation, earth infrared radiation and convection between the balloon and the external atmosphere while the internal thermal environment is composed of the internal natural convection and internal infrared radiation between the film and the internal lifting gas [28]. Therefore, the temperature differential equations of the film and membrane can be expressed by [29]

$$m_f \cdot c_f \cdot \dot{T}_f = q_D + q_S + q_R + q_{IR_atm} + q_{IR_ear} + q_{IR_He} + q_{IR_air} + q_{conv_ex} + q_{conv_in} \quad (10)$$

$$m_m \cdot c_m \cdot \dot{T}_m = q_{IR_in_he_m} - q_{IR_in_m_air} + q_{con_he_m} - q_{con_m_air} \quad (11)$$

where $m_{f/m}$, $c_{f/m}$ and $T_{f/m}$ are the mass, specific heat and temperature of the film/membrane.

The absorbed direct solar radiation energy, atmosphere scattered radiation energy and earth reflected radiation energy are given by

$$q_D = \alpha \cdot I_D \cdot A_{projected} \cdot (1 + \tau/(1 - r)) \quad (12)$$

$$q_S = \alpha \cdot I_S \cdot A_{surf} \cdot (1 + \tau/(1 - r)) \quad (13)$$

$$q_R = \alpha \cdot I_R \cdot A_{projected} \cdot (1 + \tau/(1 - r)) \quad (14)$$

The infrared radiation energy from atmosphere, earth, Helium and air are given by

$$q_{IR_atm} = \alpha_{IR} \cdot I_{IR_atm} \cdot A_{surf} \cdot (1 + \tau_{IR}/(1 - r_{IR})) \quad (15)$$

$$q_{IR_ear} = \alpha_{IR} \cdot I_{IR_ear} \cdot A_{projected} \cdot (1 + \tau_{IR}/(1 - r_{IR})) \quad (16)$$

$$q_{IR_He} = \alpha_{IR} \cdot I_{IR_He} \cdot A_{surf_He} \cdot (1/(1 - r_{IR})) \quad (17)$$

$$q_{IR_air} = \alpha_{IR} \cdot I_{IR_air} \cdot A_{surf_air} \cdot (1/(1 - r_{IR})) \quad (18)$$

where α , τ and r are the average absorptivity, transmittance and reflectivity of the film. α_{IR} , τ_{IR} and r_{IR} are the infrared absorptivity, transmittance and reflectivity of the film. A_{surf} is the surface area of the balloon. $A_{projected}$ is the top projected area [8].

The convection energy between the film and external atmosphere is described as

$$q_{conv_ex} = h_{ex} \cdot A_{surf} \cdot (T_{atm} - T_f) \quad (19)$$

The convection energy between the film and inner gas is given by

$$q_{conv_in} = q_{conv_in_He} + q_{conv_in_air} = h_{in_He} \cdot A_{surf_He} \cdot (T_{He} - T_f) + h_{in_air} \cdot A_{surf_air} \cdot (T_{air} - T_f) \quad (20)$$

where h_{ex} and h_{in} are the external and internal convective heat transfer coefficient.

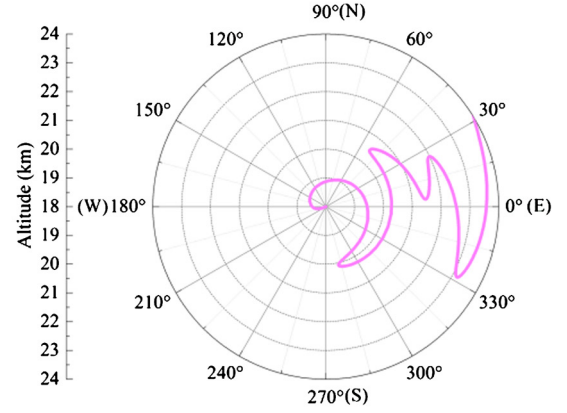
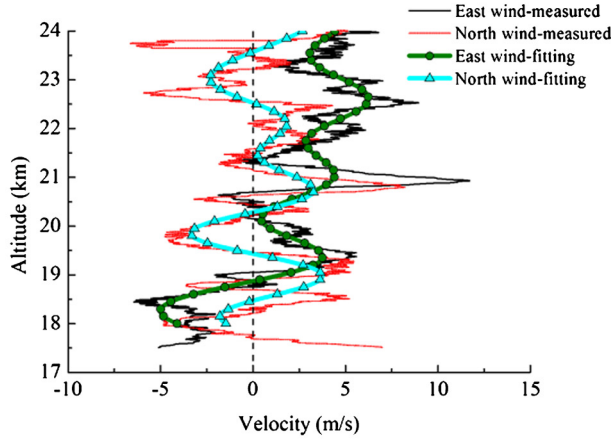
The temperature differential equations of Helium and air can be expressed by [30]

$$m_{He} \cdot c_{v_He} \cdot \dot{T}_{He} = (-q_{conv_in_He} - q_{IR_He}) + R_{He} \cdot T_{He} \cdot \dot{m}_{He} + c_{v_he} \cdot T_{He} \cdot \dot{m}_{He} - P_{He} \cdot \dot{V}_{He} \quad (21)$$

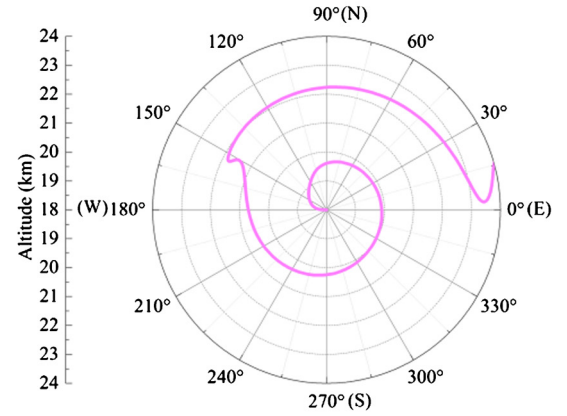
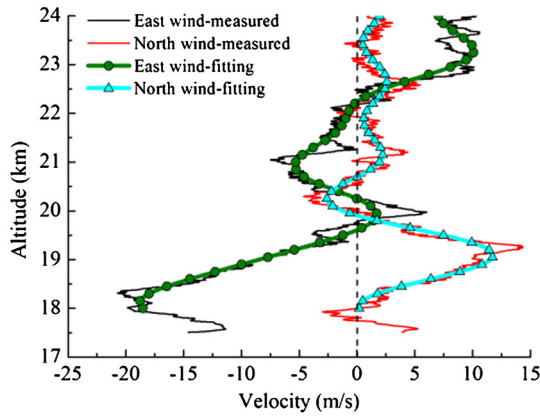
$$m_{air} \cdot c_{v_air} \cdot \dot{T}_{air} = (-q_{conv_in_air} - q_{IR_air}) + R_{air} \cdot T_{air} \cdot \dot{m}_{air} + c_{v_air} \cdot T_{air} \cdot \dot{m}_{air} - P_{air} \cdot \dot{V}_{air} \quad (22)$$

where c_v is the specific heat at constant volume of inner gas.

(a) May 15, 2018



(b) June 15, 2018



(c) July 15, 2018

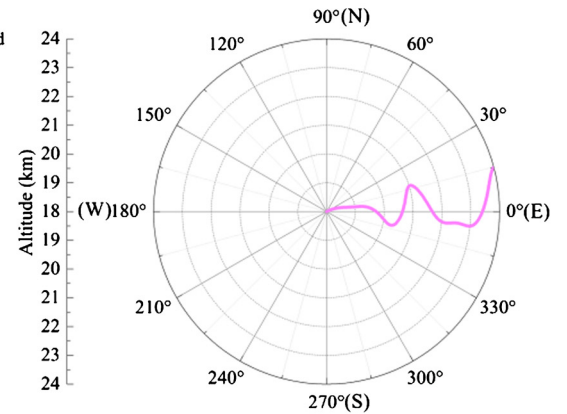
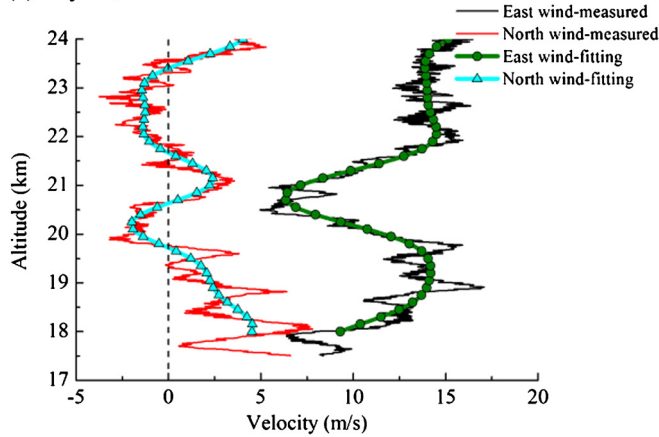


Fig. 3. Wind profiles and wind directions of Changsha on different dates.

3.3. Dynamic model

The total buoyant force and drag force produced by the balloon are given by [31]

$$B = \rho_{atm} \cdot g \cdot V \quad (23)$$

$$D = 0.5 \cdot \rho_{atm} \cdot v_r^2 \cdot C_d \cdot S \quad (24)$$

where ρ_{atm} is the density of atmosphere. C_d is the drag coefficient with a value of 0.2 [32]. S is the reference area. v_r is the relative wind velocity with respect to the movement of the balloon.

$$v_r = (v_{rx}^2 + v_{ry}^2 + v_{rh}^2)^{0.5} \quad (25)$$

where v_{rx} , v_{ry} and v_{rh} are the velocity components according to the coordinate system.

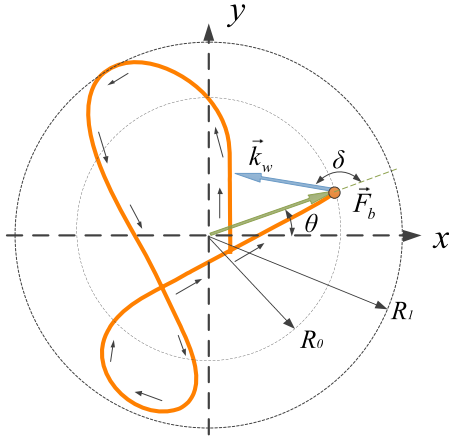


Fig. 4. Schematic of the direction transformation mechanism of the balloon.

$$\begin{cases} v_{rx} = v_{wx} - v_x \\ v_{ry} = v_{wy} - v_y \\ v_{rh} = v_{wh} - v_h \end{cases} \quad (26)$$

where v_{wx} , v_{wy} and v_{wh} are the absolute velocity components of wind, v_x , v_y and v_h are the absolute velocity components of the balloon.

Then the drag force components can be expressed by

$$\begin{cases} D_x = D \cdot v_{rx}/v_r \\ D_y = D \cdot v_{ry}/v_r \\ D_h = D \cdot v_{rh}/v_r \end{cases} \quad (27)$$

The differential equation governing the horizontal and vertical motions of the balloon is given by [33,34]

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{h} = v_h \end{cases} \quad (28)$$

$$\begin{cases} \dot{v}_x = D_x/(m + m_{add}) \\ \dot{v}_y = D_y/(m + m_{add}) \\ \dot{v}_h = (B + D_h - m \cdot g)/(m + m_{add}) \end{cases} \quad (29)$$

where x , y and h are the coordinate positions of the balloon, m is the gross mass of the balloon and m_{add} is the added mass.

$$m = m_{He} + m_{air} + m_{payload} + m_s \quad (30)$$

$$m_{add} = C_V \cdot \rho_{atm} \cdot V \quad (31)$$

where $m_{payload}$ and m_s are the mass of the payload and structure. The added mass coefficient is $C_V = 0.5$ [35].

3.4. Station-keeping model

The ACS is employed to exploit the natural wind field available in atmosphere to keep the station of the balloon within the designated district. Fig. 4 illustrates the strategy that controls the float direction of the balloon. In the designated district with a radius of R_1 , the location of the balloon changes with time. To retain the balloon within the district, a reasonable strategy is to adjust the altitude to the expected altitude where the wind direction is contrary to that of the previous altitude before the balloon floats out of the district. Considering that the balloon would continue to leave the district before it ascends or descends to the expected wind layer, the ACS should start to work a little earlier before the balloon reaches the boundary of the district. The radius where the ACS starts to work is defined as R_0 .

Table 1

Design parameters of high altitude super-pressure balloon.

Parameters	Value
Diameter, m	15.0
Volume, m ³	1767
Area, m ²	707
Gross mass, kg	155
Minimum Helium bag volume, m ³	1060.2
Maximum air ballonnet volume, m ³	706.8
Film mass, kg	56
Payload mass, kg	50
Float altitude, m	20000

The expected altitude h_{expect} can be given by

$$\max : \delta = \delta(t, h) = \arccos\left(\frac{\vec{F}_b(t) \cdot \vec{k}_w(h)}{|\vec{F}_b(t)| \cdot |\vec{k}_w(h)|}\right)$$

subject to: $x^2(t) + y^2(t) \geq R_0^2$

$$h_{\min} \leq h \leq h_{\max} \quad (32)$$

where $\vec{F}_b(t)$ is the orientation of the balloon at the moment of t , $\vec{k}_w(h)$ is the wind direction at the altitude of h , x and y are the locations of the balloon according to the coordinate system.

Then the valves will be opened to vent or pump the air to ascend or descend the balloon to the expected altitude.

$$\dot{m}_{air} = \begin{cases} C_1 & \text{if } x^2(t) + y^2(t) \geq R_0, h_{expect} < h_0 \\ 0 & \text{if } x^2(t) + y^2(t) < R_0 \\ -C_2 & \text{if } x^2(t) + y^2(t) \geq R_0, h_{expect} > h_0 \end{cases} \quad (33)$$

where C_1 and C_2 are constants ($C_1 > 0, C_2 > 0$). h_0 is the altitude of the balloon when the balloon floats out of the district with the radius of R_0 .

The flow chart of the station-keeping program for the balloon is shown in Fig. 5.

4. Simulation results and analysis

4.1. Simulation condition

Taking the high altitude balloon with a diameter of 15.0 m as an example, the main design parameters of the balloon are provided in Table 1. The maximum air ballonnet volume is defined as the volume of air ballonnet when the differential pressure reaches the maximum value ($\Delta P_{\max}$). And the minimum Helium bag volume is defined as the corresponding volume of Helium bag at this moment.

4.2. Results and analysis

4.2.1. Altitude control performance

To illustrate the availability of the ACS, the variations of the air mass, velocity and altitude during the venting and pumping processes are simulated and shown in Fig. 6 and Fig. 7. In order to describe the processes completely, we assume that the exhaust/aeration valves are opened at the moment of 1.0 hour.

During the venting process, the air mass decreases from 45.0 kg to 21.6 kg within 1.08 hours with a rate of -6.0 g/s. Accordingly, the ascent velocity increases to 0.34 m/s within several minutes and then stabilizes at about 0.24 m/s. Then it is noteworthy that a slight increase of the velocity appears. This can be explained by the nonlinear decrease of the atmosphere density. About 1.08 hours later, the velocity declines and converges to zero eventually. As a result, the altitude ascends to 21.0 km from 20.0 km. On the contrary, the pumping rate of the air is set as 4 g/s during the

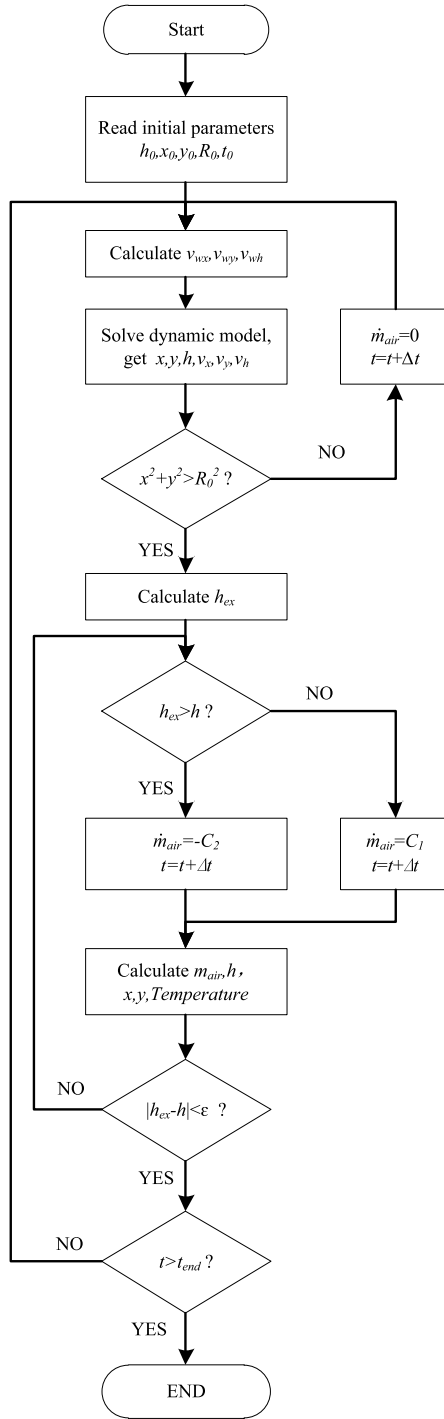


Fig. 5. Flow chart of the station-keeping program for the balloon.

pumping process. Therefore, it takes about 1.84 hours to increase the air mass from 45.0 kg to 71.4 kg. Caused by the growth of the density of the balloon, the altitude descends to 19.0 km from 20.0 km with the velocity of about -0.15 m/s. As a conclusion, the results above state that the ACS proposed in this paper is available to adjust the hover altitude of the balloon.

4.2.2. Station-keeping performance

Assuming that the wind profiles are steady in several days for simplicity, the trajectory of the balloon with the ACS within 24 hours in May 15, 2018 is shown in Fig. 8. It can be seen that the cruise direction changes whenever the balloon floats out of the

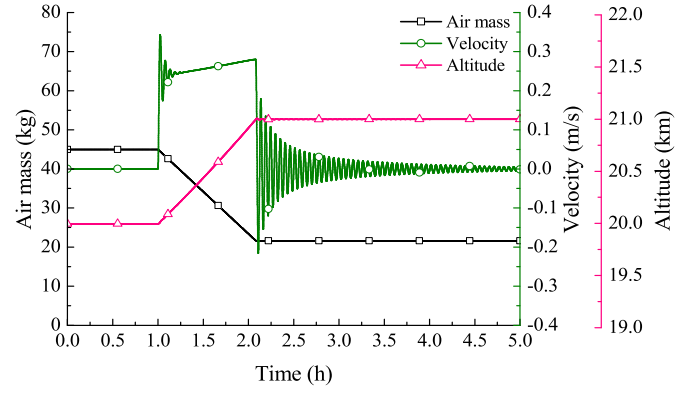


Fig. 6. Variations of air mass, velocity and altitude during venting process.

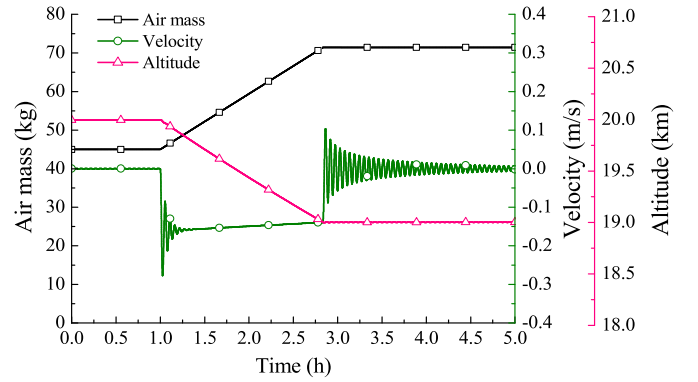


Fig. 7. Variations of air mass, velocity and altitude during pumping process.

designated district. As a result, the balloon is limited in the district that with a radius of about 25 km. The change of the cruise direction is realized via the ACS. By charging the air from atmosphere into the air ballonet or the reversed process, the hover altitude can be adjusted to utilize the wind with different directions at different altitudes. Fig. 9 indicates the changes of the air mass, altitude and differential pressure between inner gas and atmosphere during the venting and pumping processes within 24 hours. It can be seen that the air mass increases from 45 kg to 84 kg during the first pumping process that lasts from 00:50 am to 03:30 am. Consequently, the float altitude descends from 19.5 km where the wind direction is 346.5° (Fig. 3(a)) to 18.2 km where the wind direction is 199.2° (Fig. 3(a)). It is obvious that the wind direction is reversed. The differential pressure between inner gas and atmosphere increases from about 400 Pa to about 500 Pa. Then the first venting process occurs and lasts from 03:30 am to 03:50 am. During this process, the air mass decreases from 84 kg to 79 kg and the altitude ascends about 150 m. After a transient equilibrium, the second pumping process occurs and lasts from 04:15 am to 06:30 am. The air mass decreases to only about 30 kg and the altitude ascends to 20.1 km where the wind direction is 282.4° (Fig. 3(a)). Subsequently, several pumping and venting processes appear for the rest of the day.

The processes of venting and pumping gas have direct influences on the temperatures of the internal gases for the expansion and compression effects. Therefore, we calculate the temperature fluctuations of Helium and air in the balloon caused by venting and pumping processes. The results are given in Fig. 10. As a comparison, the Helium temperature of the balloon without ACS is also calculated. It can be seen that the Helium and air temperatures increase and decrease obviously during the venting process and pumping process respectively. At the beginning of the first pumping process, the air temperature and Helium temperature increase

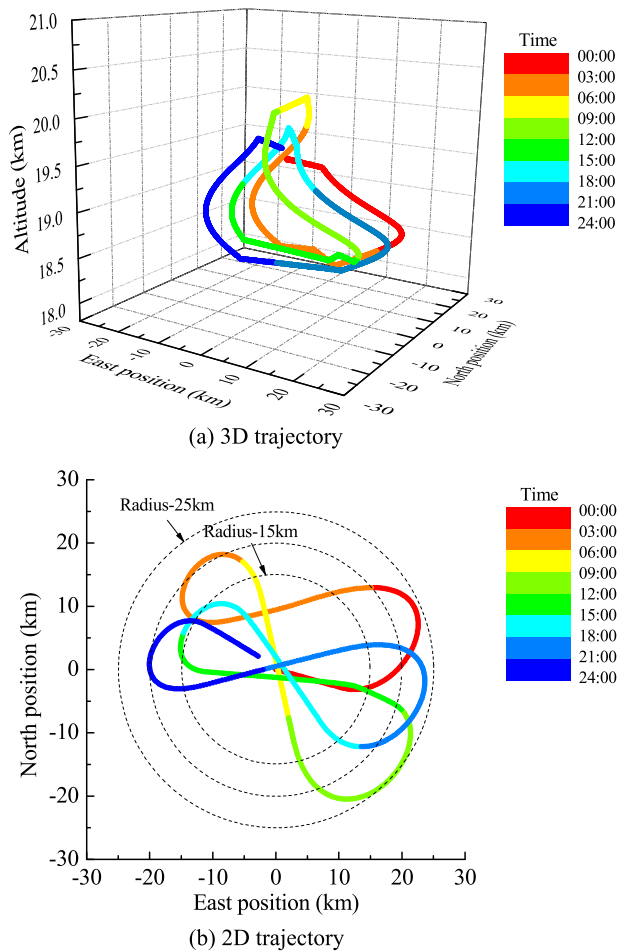


Fig. 8. Station-keeping performance of the high altitude balloon with altitude control system in May 15, 2018.

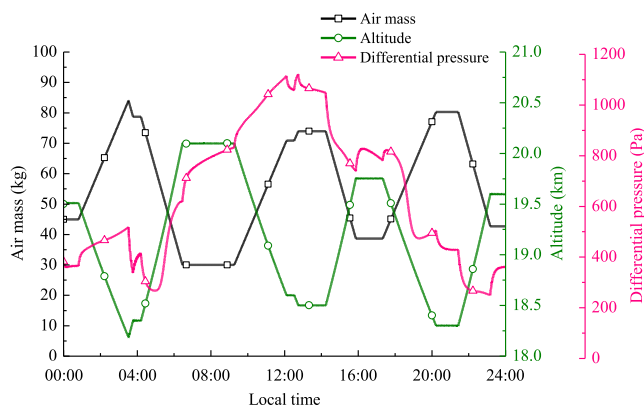


Fig. 9. Variations of air mass, altitude and differential pressure of the balloon in May 15, 2018.

about 4 K and 1 K within 20 minutes respectively for the compression effect. After that, the temperatures no longer increase in spite of the lasting of the pumping process for more than two hours. This may be contributed by the enhancement of the heat convection and the infrared radiation among the air, the Helium and the atmosphere which is proportional to the temperature difference. For the same reason, the temperatures of air and Helium decrease about 7 K and 3 K respectively within 30 minutes and then keep constant during the second venting process until the sun rises and heats the inner gas. At noon, the maximum temperature of Helium

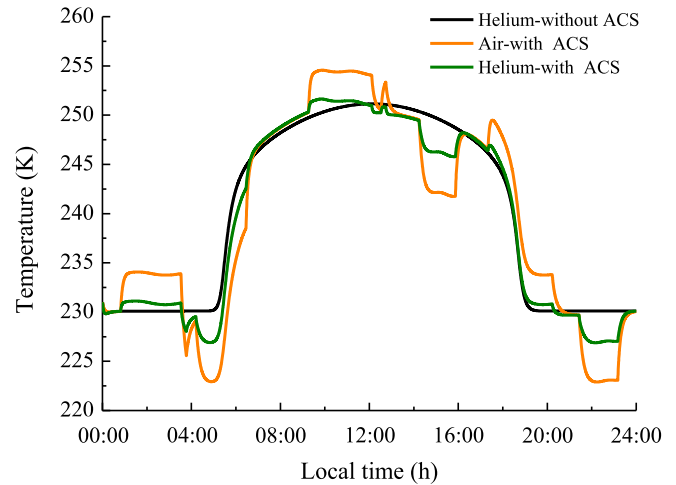


Fig. 10. Variations of the temperature of air and Helium within May 15, 2018.

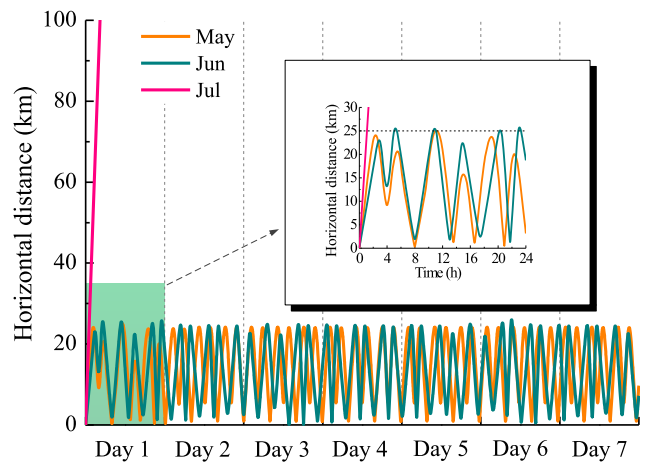


Fig. 11. Variation of the horizontal distance of the balloon.

is about 251 K without the venting/pumping process while the air temperature increases to about 255 K when the air is pumped into the air ballonet.

Fig. 11 displays the change of the horizontal distance between the balloon and the center of the designated district within seven days in different wind fields. It can be seen that the distances are less than 25 km in the wind field of May and Jun, which means that the balloon with the ACS can divert the course by using wind currents and keep station in the designated district. The trajectories of the balloon in 1st day and 7th day are revealed in Fig. 12. It can be seen that the trajectories are chaotic in the 1st day. However, for the reason that the wind fields are assumed to be relatively stable in several days, the trajectories tend to be stable in the 7th day. Considering that the real wind field is time-varying, the real trajectory of the balloon should be irregular and the stable situation is very unlikely [36]. However, it can still prove that the ACS developed in this paper is available to keep the balloon in the designated district for several days.

It can also be seen that the distance grows swiftly to more than 100 km within several hours in the wind field of July in Fig. 11. This can be explained by the characteristic of the wind field in July (Fig. 3(c)). The wind is easterly and the direction changes slightly from the altitude of 18 km to 24 km. As a result, the ACS is disabled to reverse the float direction by changing the cruise altitude. This phenomenon indicates that the ACS proposed in this paper strongly depends on the characteristic of wind field. Only in the wind field that the direction changes greatly with the altitude,

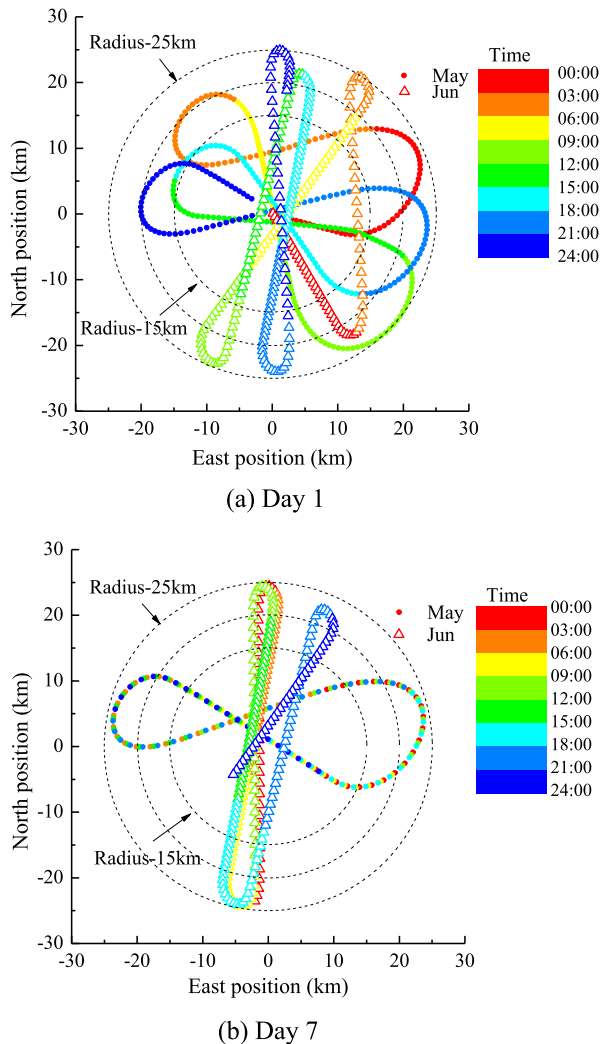


Fig. 12. 2D trajectories of the balloon at Day 1 and Day 7.

the ACS is feasible to keep the station of the balloon. Nevertheless, the reverse wind exists for some time of the year, but not all times of the year [21]. For the wind field that the direction is approximately constant, an effective approach to accomplish the communication relay mission is, referring to the GPS system, employing multi-balloons system which will be researched next.

5. Conclusions

To fix the high altitude balloon within designated district, the altitude control system, which can adjust the float altitude and then change the cruise direction by exploiting the natural wind field, is proposed in this paper. Based on the theoretical models of the real wind field model, thermal model, dynamic model and station-keeping model, a Matlab program is developed to numerically investigate the station-keeping performance of the balloon. The following conclusions are drawn.

- (1) The altitude control system can adjust the float altitude of the balloon successfully by venting air to the atmosphere or pumping air into the air balloonet.
- (2) The temperatures of the air and Helium fluctuate only several Kelvins during the venting and pumping processes. The influence of the fluctuation on the flight performance is negligible.
- (3) By adjusting the float altitude after identifying the wind layer with the desired direction, the balloon can exploit the wind to

keep the station within the target district in real wind fields for several days.

In this paper, we assume that the wind field is relative stable in several days. However, the real wind field is time-varying in reality. Hence, the future research can be directed towards the station-keeping performance of the balloon in the time-varying real wind field. In addition, for the wind field that the wind direction changes slightly with altitude, the altitude control system is unserviceable to keep station. Therefore, the multi-balloons distributed control system should be researched next to provide persistent supervision in the target district.

Declaration of Competing Interest

There is no competing interest.

Acknowledgements

This work was supported by the Academic Excellence Foundation of BUAA for PHD Students (Grant No. 201905034), National Natural Science Foundation of China under Grant Nos. 51307004, 51775021 and 11602004, National Key Research and Development Program of China (2016YFB0500801) and Innovation-Driven Project of Central South University (No. 2018CX024).

References

- [1] Huafei Du, Weiyu Zhu, Yifei Wu, et al., Effect of angular losses on the output performance of solar array on long-endurance stratospheric airship, *Energy Convers. Manag.* 147 (2017) 135–144, <https://doi.org/10.1016/j.enconman.2017.05.047>.
- [2] Jianhui Hu, Yipo Li, Wujun Chen, et al., Uniaxial mechanical properties of multi-layer thin films in use for scientific balloons, *Adv. Space Res.* 62 (5) (2018) 1165–1176, <https://doi.org/10.1016/j.asr.2018.06.003>.
- [3] Nobuyuki Yajima, Takeshi Imamura, Naoki Izutsu, et al., The future of scientific ballooning, in: Abe, et al. (Eds.), *Scientific Ballooning: Technology and Applications of Exploration Balloons Floating in the Stratosphere and the Atmospheres of Other Planets*, Springer, New York, NY, 2009, pp. 195–203.
- [4] M.S. Smith, E.L. Rainwater, Optimum designs for superpressure balloons, *Adv. Space Res.* 33 (10) (2004) 1688–1693, <https://doi.org/10.1016/j.asr.2003.07.042>.
- [5] Jihui Qiu, Shaoping Shen, Zhibin Li, Station-keeping control for a stratosphere airship via wind speed prediction approach, *Int. J. Intell. Comput. Cybern.* 10 (4) (2017), <https://doi.org/10.1108/IJICC-02-2017-0017>.
- [6] András Söbester, Helen Czerski, Niccolò Zapponi, et al., High-altitude gas balloon trajectory prediction: a Monte Carlo model, *AIAA J.* 52 (4) (2014) 832–842, <https://doi.org/10.2514/1.J052900>.
- [7] Valerian Jewtoukoff, Riwal Plougonven, Albert Hertzog, et al., On the prediction of stratospheric balloon trajectories: improving winds with mesoscale simulations, *J. Atmos. Ocean. Technol.* 33 (8) (2016) 1629–1647, <https://doi.org/10.1175/jtech-d-15-0110.1>.
- [8] Huafei Du, Jun Li, Weiyu Zhu, et al., Flight performance simulation and station-keeping endurance analysis for stratospheric super-pressure balloon in real wind field, *Aerosp. Sci. Technol.* 86 (2019) 1–10, <https://doi.org/10.1016/j.ast.2019.01.001>.
- [9] Henry M. Cathey, Debora A. Fairbrother, Magdi A. Said, Performance highlights of NASA super pressure balloon mid-latitude flights, in: *AIAA Balloon Systems Conference, American Institute of Aeronautics and Astronautics*, 2017.
- [10] Henry Cathey, David Pierce, Duration flight of the NASA super pressure balloon, in: *AIAA Balloon Systems Conference, American Institute of Aeronautics and Astronautics*, 2009.
- [11] Jun Li, Jun Liao, Yuxin Liao, et al., An approach for estimating perpetual endurance of the stratospheric solar-powered platform, *Aerosp. Sci. Technol.* 79 (2018) 118–130, <https://doi.org/10.1016/j.ast.2018.05.035>.
- [12] Yueneng Yang, Positioning control for stratospheric satellites subject to dynamics uncertainty and input constraints, *Aerosp. Sci. Technol.* 86 (2019) 534–541, <https://doi.org/10.1016/j.ast.2019.01.045>.
- [13] J. Morgado, M. Abdollahzadeh, M.A.R. Silvestre, et al., High altitude propeller design and analysis, *Aerosp. Sci. Technol.* 45 (2015) 398–407, <https://doi.org/10.1016/j.ast.2015.06.011>.
- [14] Durlabh Pande, Dries Verstraete, Impact of solar cell characteristics and operating conditions on the sizing of a solar powered nonrigid airship, *Aerosp. Sci. Technol.* 72 (2018) 353–363, <https://doi.org/10.1016/j.ast.2017.11.013>.

- [15] Sai Sudha Ramesh, Kian Meng Lim, Heow Pueh Lee, et al., Reduced-order modeling of stratospheric winds and its application in high-altitude balloon trajectory simulations, *J. Appl. Meteorol. Climatol.* 56 (6) (2017) 1753–1766, <https://doi.org/10.1175/JAMC-D-16-0294.1>.
- [16] Kerry T. Nock, Kim M. Aaron, Matthew K. Heun, et al., Aerodynamic and mission performance of a winged balloon guidance system, *J. Aircr.* 44 (6) (2007) 1923–1938, <https://doi.org/10.2514/1.31922>.
- [17] Christopher D. Yoder, Thomas R. Gemmer, Andre P. Mazzoleni, Modelling and performance analysis of a tether and sail-based trajectory control system for extra-terrestrial scientific balloon missions, *Acta Astronaut.* 160 (2019) 527–537, <https://doi.org/10.1016/j.actaastro.2018.12.030>.
- [18] Sai Sudha Ramesh, Juanli Ma, Kian-Meng Lim, et al., Numerical evaluation of station-keeping strategies for stratospheric balloons, *Aerosp. Sci. Technol.* 80 (2018) 288–300, <https://doi.org/10.1016/j.ast.2018.07.010>.
- [19] Paul B. Voss, Emily E. Riddle, Mike Smith, Altitude control of long-duration balloons, *J. Aircr.* 42 (2) (2005) 478–482, <https://doi.org/10.2514/1.7481>.
- [20] Rafy U. Athar, Taylor E. Mathews, Jessica M. Lavigne, et al., Stratospheric C4ISR unmanned station (STRATACUS), in: *AIAA Balloon Systems Conference*, American Institute of Aeronautics and Astronautics, 2017.
- [21] Michael S. Smith, Demonstration of Fine Altitude Control on Stratospheric Balloons to Achieve a Desired Ground Track, 2017.
- [22] Jeffery L. Hall, Jonathan Cameron, Michael Pauken, et al., Altitude-controlled light gas balloons for venus and titan exploration, in: *AIAA Aviation 2019 Forum*, American Institute of Aeronautics and Astronautics, 2019.
- [23] E. Mack, Balloon-powered internet for everyone, Project Loon [Online]. Available: <https://x.company/loon/>.
- [24] K. Kamnani, C. Suratkhar, A review paper on Google Loon Technique, *Int. J. Res. Sci.* 1 (1) (2013) 167–171.
- [25] J.E. Laby, E.L. Unthank, Day-to-day fluctuations of the winds in the lower stratosphere over Australia, *J. Atmos. Sci.* 24 (1966) 98–100.
- [26] Weiyu Zhu, Jun Li, Yuanming Xu, Optimum attitude planning of near-space solar powered airship, *Aerosp. Sci. Technol.* 84 (2019) 291–305, <https://doi.org/10.1016/j.ast.2018.10.007>.
- [27] Kazuhisa Chiba, Ryosuke Nishikawa, Masahiko Onda, et al., Aerodynamic influences on a tethered high-altitude lighter-than-air platform system to its behavior, *Aerosp. Sci. Technol.* 70 (2017) 405–411, <https://doi.org/10.1016/j.ast.2017.08.029>.
- [28] Huafei Du, Jun Li, Weiyu Zhu, et al., Thermal performance analysis and comparison of stratospheric airships with rotatable and fixed photovoltaic array, *Energy Convers. Manag.* 158 (2018) 373–386, <https://doi.org/10.1016/j.enconman.2017.12.078>.
- [29] Arturo González-Llana, David González-Bárcena, Isabel Pérez-Grande, et al., Selection of extreme environmental conditions, albedo coefficient and Earth infrared radiation, for polar summer Long Duration Balloon missions, *Acta Astronaut.* 148 (2018) 276–284, <https://doi.org/10.1016/j.actaastro.2018.05.016>.
- [30] Xixiang Yang, Prediction of thermal behavior and trajectory of stratospheric airships during ascent based on simulation, *Adv. Space Res.* 57 (11) (2016) 2326–2336, <https://doi.org/10.1016/j.asr.2016.02.030>.
- [31] Qiumin Dai, Xiande Fang, Xiaojian Li, et al., Performance simulation of high altitude scientific balloons, *Adv. Space Res.* 49 (6) (2012) 1045–1052, <https://doi.org/10.1016/j.asr.2011.12.026>.
- [32] Jaber Almedej, Drag coefficient of flow around a sphere: matching asymptotically the wide trend, *Powder Technol.* 186 (3) (2008) 218–223, <https://doi.org/10.1016/j.powtec.2007.12.006>.
- [33] Roberto Palumbo, Michelangelo Russo, Edoardo Filippone, et al., ACHAB: analysis code for high-altitude balloons, in: *AIAA Atmospheric Flight Mechanics Conference and Exhibit*, American Institute of Aeronautics and Astronautics, 2007.
- [34] A. Ceruti, D. Gambacorta, P. Marzocca, Unconventional hybrid airships design optimization accounting for added masses, *Aerosp. Sci. Technol.* 72 (2018) 164–173, <https://doi.org/10.1016/j.ast.2017.10.042>.
- [35] Yi Zhang, Dongxu Liu, Influences of initial launch conditions on flight performance of high altitude balloon ascending process, *Adv. Space Res.* 56 (4) (2015) 605–618, <https://doi.org/10.1016/j.asr.2015.04.031>.
- [36] Selvaraj Dharmalingam, Riwal Plougonven, Albert Hertzog, et al., Accuracy of balloon trajectory forecasts in the lower stratosphere, *Atmosphere* 10 (2) (2019), <https://doi.org/10.3390/atmos10020102>.